



Chronos DEX

Team Jelly Fish: Nihal Pandey &
Anshika Srivastava



Chronos **DEX!**

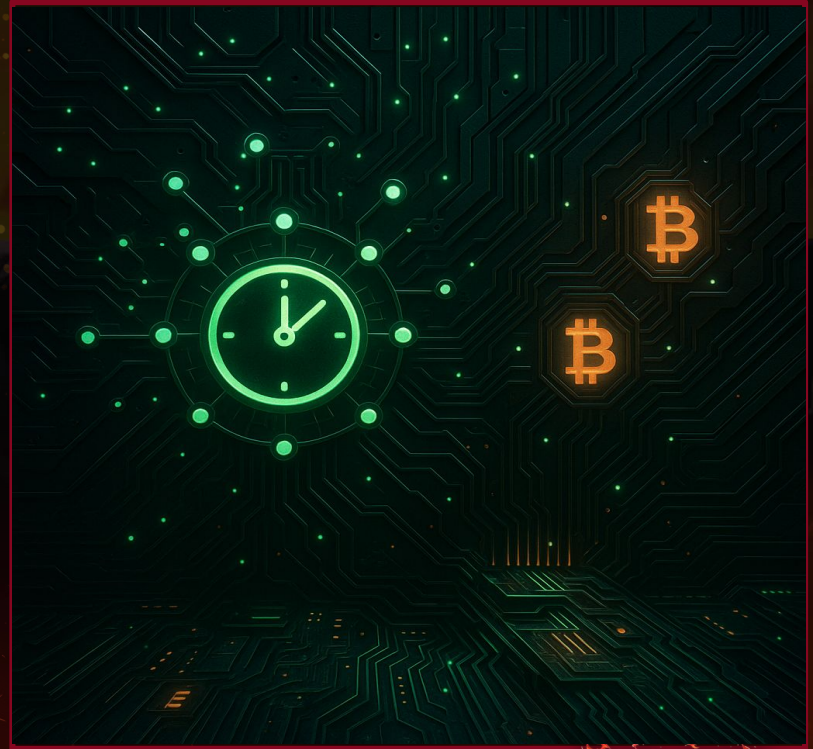
First Layla-Ready AMM on
Bitcoin Cash

<https://github.com/Nihal-Pandey-2302/Chronos-Dex>

The Story of Chronos DEX

The BCH BLAZE hackathon challenged us to build with the new Layla upgrade CHIPS. We asked: "What if a DEX could not only swap tokens efficiently, but also GAMIFY trading to make DeFi more engaging?"

We combined AMM mechanics with Battle.cash integration and used EVERY major Layla feature to create Chronos DEX - where traders earn XP on-chain for every swap!

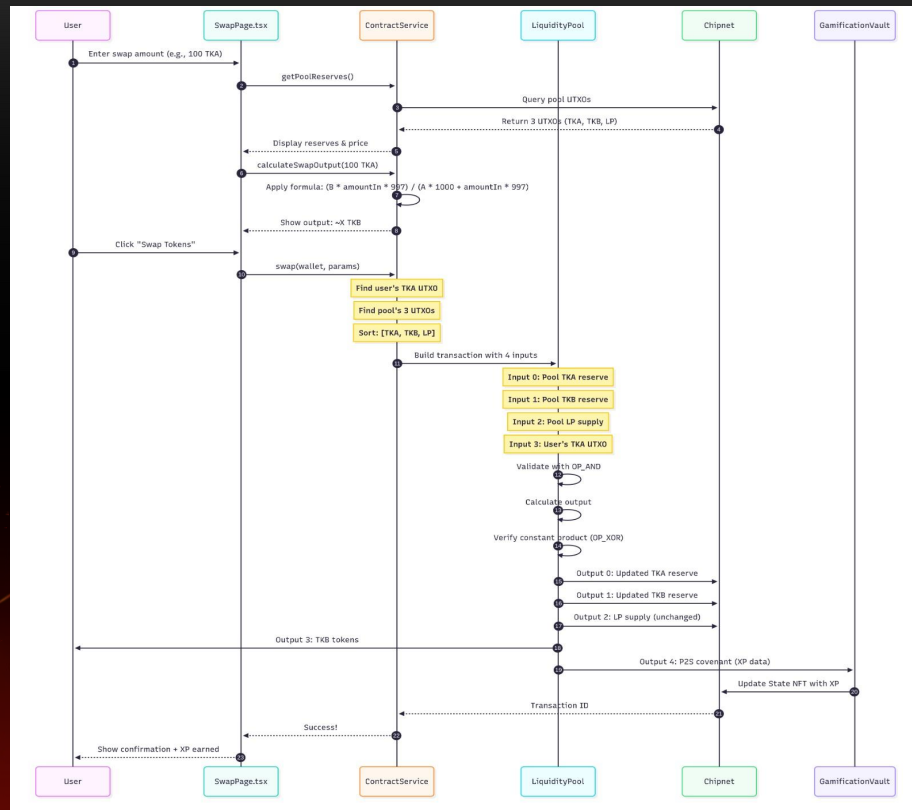
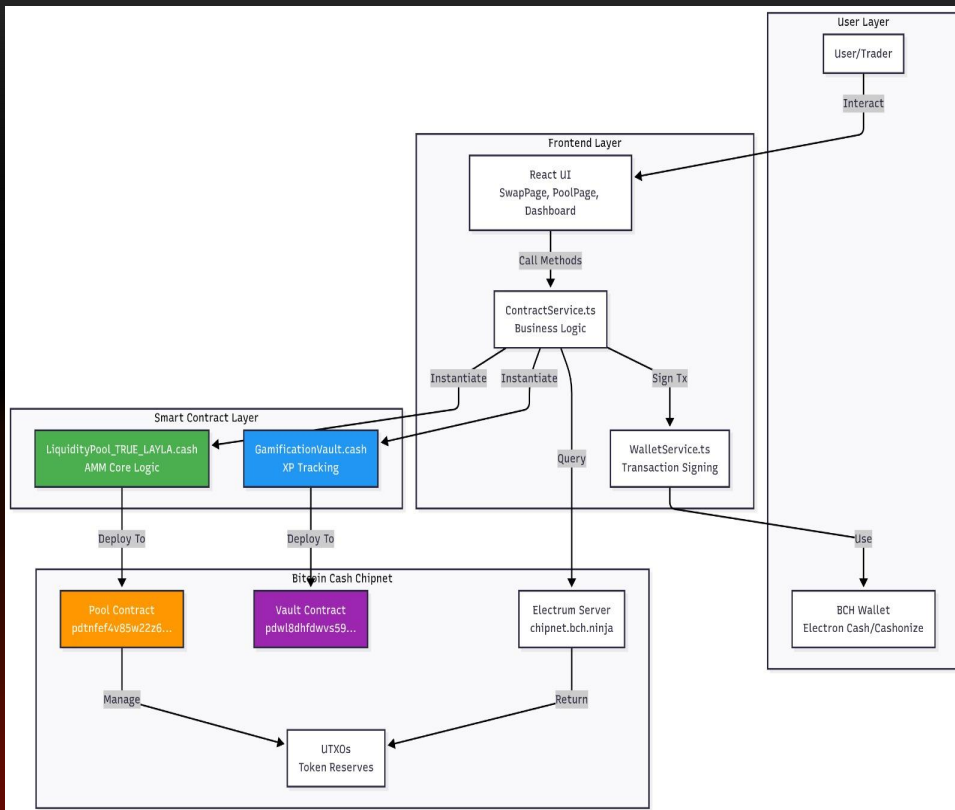


Benefits of Chronos DEX

- **Swap tokens with Layla efficiency**
 - Bitwise operations = 10x cheaper validation
 - P2S outputs enable trustless XP tracking
 - Composite tokens prove liquidity provision
- **Your trading history lives on-chain forever**
 - Every swap creates a Battle.cash P2S output
 - XP is permanently recorded and verifiable
- **10x more engaging than traditional AMMs**
 - Leaderboards, achievements, gamification
 - Battle.cash integration makes DeFi fun!



Architecture Diagram



Layla Example

Layla CHIPs Used

- ✓ BITWISE (10M bounty) - AND/OR operations for validation flags - XOR to verify constant product changed - Lines 55-67, 120-126 in LiquidityPool.cash
- ✓ P2S (10M bounty) - Output 4 in swap() creates P2S covenant - Enables Battle.cash XP tracking - Line 196-200 in LiquidityPool.cash
- ✓ COMPOSITE TOKENS (100M bounty) - LP tokens = FT (amount) + NFT (receipt) - Output 3 in addLiquidity() - Lines 98-103 in LiquidityPool.cash
- ✓ BATTLE.CASH (100M bounty) - GamificationVault.cash processes P2S - On-chain leaderboard via State NFT - Trustless XP accumulation

Layla Usage

```

// 🌀 LAYLA OP_AND: Validate amountIn is positive
bytes mask = 0x7FFFFFFF;
require(amountIn == int(bytes(amountIn) & mask));

// 🌀 LAYLA OP_OR: Combine validation flags
int validationFlags = 0;
if (lpTokensToMint > 0) {
    validationFlags = validationFlags | 1; // Set bit 0
}
if (amountA > 0) {
    validationFlags = validationFlags | 2; // Set bit 1
}
require((validationFlags & 3) == 3); // Both must be set

// 🌀 LAYLA OP_XOR: Verify constant product increased
require((newReserveA * newReserveB) ^ (reserveA * reserveB) > 0);

// 🌀 P2S COVENANT: Enable trustless XP tracking
bytes swapData = userPkh + bytes8(amountIn) + poolId;
require(tx.outputs[4].nftCommitment == swapData);
require(tx.outputs[4].value >= 1000);

// 🌀 COMPOSITE TOKENS: LP tokens with NFT receipt
bytes commitment = bytes4(int(tx.locktime)) + bytes4(0);
require(tx.outputs[3].tokenAmount == lpTokensToMint); // Fungible
require(tx.outputs[3].nftCommitment == commitment); // Non-fungible
```


The Vision for Chronos DEX

- Now that Layla enables efficient DeFi primitives, it's time to make them **ENGAGING** for users
 - Gamification → More daily active users
 - Battle.cash → Trading becomes competitive
 - On-chain reputation → Trustless credibility
- Next priority → Multi-hop routing, liquidity mining rewards, mobile app
- Target: 10,000 gamified swaps per day by 2026.



THANK YOU!

